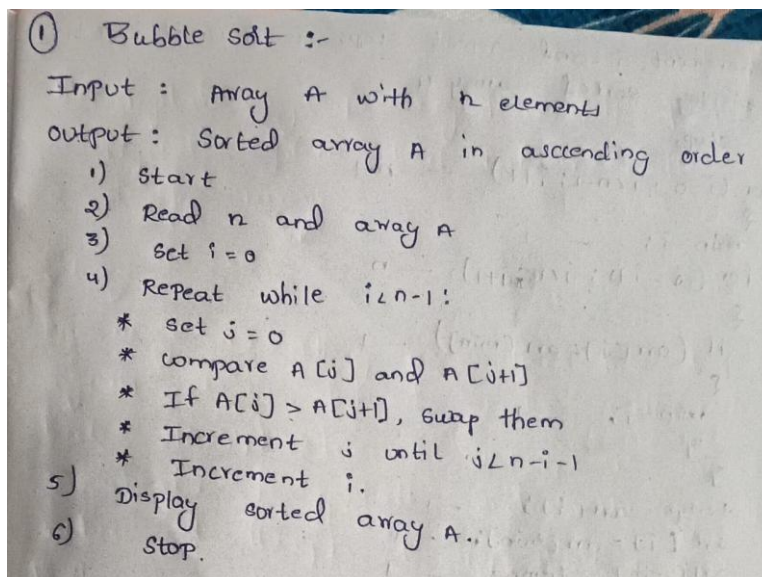


## PRACTICAL - 1

**AIM : Implementation and Time analysis of sorting algorithms. Bubble sort, Selection sort, Insertion sort, Merge sort and Quicksort**

### BUBBLE SORT ALGORITHM



PROGRAM:

```
#include <iostream>
using namespace std;
class bubblesort {
public:
    int size, arr[100], i;
    void arrayinput()

.... {      cout << "Enter array size: ";
cin >> size;      cout << "Enter array
elements:\n";      for (i = 0; i < size;
i++)
    {      cin
>> arr[i];
    }
```



```
    }    void
printarray()
{
    for (i = 0; i < size; i++)
    {
        cout << arr[i] << " ";
    }    cout
<< endl;
}
void bubblelogic()
{    int j, temp;    for
(i = 0; i < size - 1; i++)
{    for (j = 0; j < size - i -
1; j++)
{
    if (arr[j] > arr[j + 1])
    {
temp = arr[j];
arr[j] = arr[j + 1];    arr[j
+ 1] = temp;
    }
    }
}
} }; int
main() {
    bubblesort bs;
bs.arrayinput();    cout <<
"Before sorting: ";
bs.printarray();
bs.bubblelogic();    cout <<
"After sorting: ";
bs.printarray();    return 0;
}
```

OUTPUT :

```

Enter array size: 4
Enter array elements:
5
4
6
9
Before sorting: 5 4 6 9
After sorting: 4 5 6 9

...Program finished with exit code 0
Press ENTER to exit console.

```

TIME COMPLEXITY : Step Count / Frequency Count Method )

best case :

best case :  
Bubble sort  
first Pass  $\rightarrow n-1$   
no swapping  
loops stops  
 $T(n) = n-1$   
Best case =  $O(n)$

average case :

```

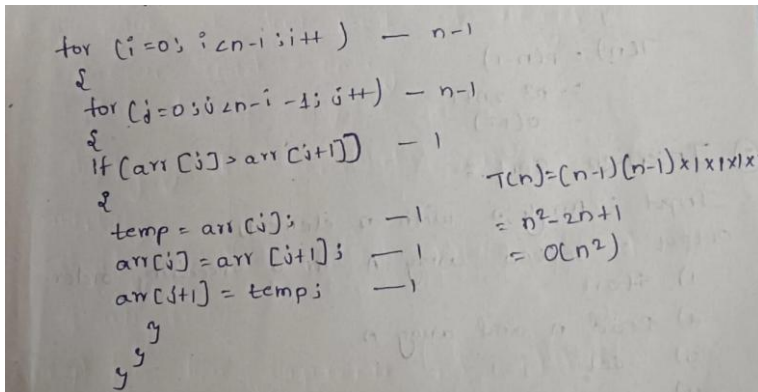
for (i=0; i<n-1; i++) — n-1
{
    for (j=0; j<n-i-1; j++) — n-1
    {
        if (arr[j] > arr[j+1]) — 1
        {
            temp = arr[j]; — 1
            arr[j] = arr[j+1]; — 1
            arr[j+1] = temp; — 1
        }
    }
}

```

$T(n) = (n-1)(n-1) \times 1 \times 1 \times 1$   
 $= n^2 - 2n + 1$   
 $= O(n^2)$



Worst case :



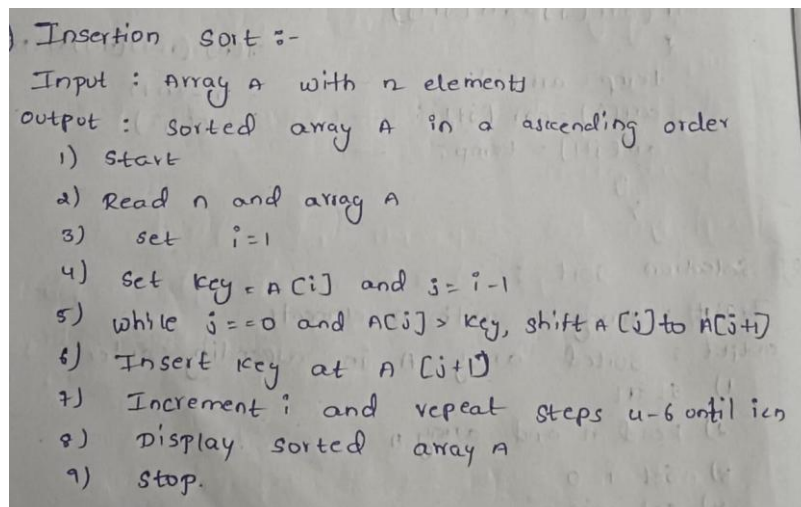
```

for (i=0; i<n-1; i++) — n-1
{
    for (j=0; j<n-i-1; j++) — n-1
    {
        if (arr[j] > arr[j+1]) — 1
        {
            temp = arr[j]; — 1
            arr[j] = arr[j+1]; — 1
            arr[j+1] = temp; — 1
        }
    }
}

```

$T(n) = (n-1)(n-1) \times 1 \times 1 \times 1$   
 $= n^2 - 2n + 1$   
 $= O(n^2)$

## INSERTION ALGORITHM



Insertion Sort :-  
 Input : Array A with n elements  
 Output : Sorted array A in a ascending order

- 1) Start
- 2) Read n and array A
- 3) set i=1
- 4) set key = A[i] and j = i-1
- 5) while j >= 0 and A[j] > key, shift A[j] to A[j+1]
- 6) Insert key at A[j+1]
- 7) Increment i and repeat steps 4-6 until i=n
- 8) Display sorted array A
- 9) Stop.

## PROGRAM:

```

#include <iostream> using
namespace std;

class InsertionSort { public:
int size, arr[100], i, j, temp;

    void arrayinput()
    {
        cout << "Enter array
size: ";    cin >> size;
    }
}

```



```
        cout << "Enter array elements: ";
for (i = 0; i < size; i++)

    {
        cin
>> arr[i];
    }

void printarray()
{

    for (i = 0; i < size; i++)
    {
        cout <<
arr[i] << " ";
    }
    cout
<< endl;
}

void insertionsortlogic()
{
    for (i = 1; i < size;
i++)
    {
        temp
= arr[i];
        j = i - 1;

        while (j >= 0 && arr[j] > temp)
        {
            arr[j +
1] = arr[j];
            j--;
        }

        arr[j + 1] = temp;
    }
}

int main()
{
    InsertionSort is;

    is.arrayinput();
```

---

```
cout << "Before sorting: ";  
is.printarray();
```

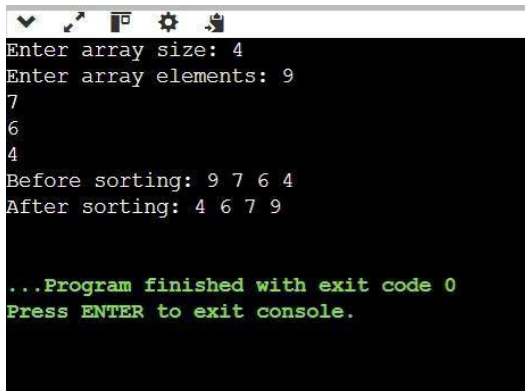
```
is.insertionsortlogic();
```

```
cout << "After sorting: ";  
is.printarray();
```

```
return 0;
```

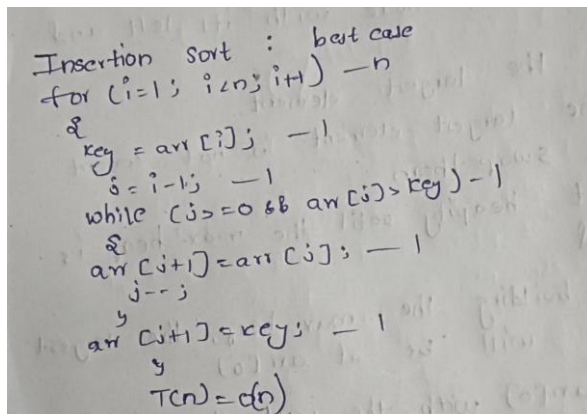
```
}
```

OUTPUT :



```
Enter array size: 4  
Enter array elements: 9  
7  
6  
4  
Before sorting: 9 7 6 4  
After sorting: 4 6 7 9  
  
...Program finished with exit code 0  
Press ENTER to exit console.
```

TIME COMPLEXITY : Step Count / Frequency Count Method )  
best case :



Insertion sort : best case  
for ( $i=1; i \leq n; i++$ ) —  $n$   
     $key = arr[i];$  — 1  
     $j = i-1;$  — 1  
    while ( $j > 0 \ \&\& \ arr[j] > key$ ) — 1  
         $arr[j+1] = arr[j];$  — 1  
         $j--;$   
     $arr[j+1] = key;$  — 1  
     $T(n) = O(n)$

average case

Average case :

Pass-1  $\rightarrow n-1$  comparisons  
Pass-2  $\rightarrow n-2$  comparisons  
 $\therefore T(n) = \frac{n(n-1)}{2}$   
ignore constants  
 $= O(n^2)$

worst case :

Worst case : it occurs when  
pivot is always smallest (&)  
& largest  
 $n + (n-1) + (n-2) + \dots + 1$   
 $T(n) = \frac{n(n+1)}{2}$   
 $= O(n^2)$

### SELECTION SORT ALGORITHM

Selection Sort :-  
Input : Array A with n elements  
Output : Sorted array A in ascending order

- 1) start
- 2) Read n and array A
- 3) Set  $i = 0$
- 4) set  $min = i$
- 5) compare A[min] with the remaining elements
- 6) If  $A[j] < A[min]$ , set  $min = j$
- 7) Swap A[i] and A[min]
- 8) Increment i and repeat steps 4-7 until  $i < n-1$ .
- 9) Display sorted array A
- 10) stop.



PROGRAM:

```
#include <iostream> using
namespace std;

class SelectionSort { public:
int size, arr[100], i, j, temp;

    void arrayinput()
    {      cout << "Enter array
size: ";      cin >> size;

        cout << "Enter array elements: ";
for (i = 0; i < size; i++)
    {      cin
>> arr[i];
    }
    }

    void printarray()
    {      for (i = 0; i < size;
i++)
        {      cout <<
arr[i] << " ";
        }      cout
<< endl;
    }

    void selectionsortlogic()
    {      for (i = 0; i < size - 1;
i++)
        {      int
min = i;

            for (j = i + 1; j < size; j++)
            {      if (arr[j]
< arr[min])      {
min = j;
            }
        }
    }
```

---



```
        }
        if (min != i)
        {
            temp =
arr[i];
arr[i] = arr[min];    arr[min]
= temp;
        }
    }
}
};

int main()
{
    SelectionSort ss;

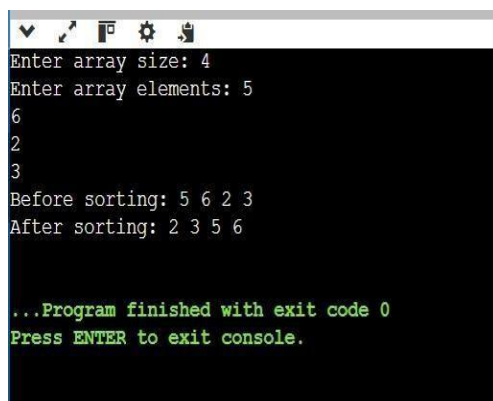
    ss.arrayinput();

    cout << "Before sorting: ";    ss.printarray();
ss.selectionsortlogic();

    cout << "After sorting: ";
ss.printarray();

    return 0;
}
```

OUTPUT :



```
Enter array size: 4
Enter array elements: 5
6
2
3
Before sorting: 5 6 2 3
After sorting: 2 3 5 6

...Program finished with exit code 0
Press ENTER to exit console.
```

time complexity : Step Count / Frequency Count Method )

best case :

```
for (i=0; i<n-1; i++) — n-1
{
    min=i; — 1
    for (j=i+1; j<n; j++) — n
    {
        if (arr[j]<arr[min]) — 1
        {
            min=j; — 1
        }
        y
    }
    temp = arr[i]; — 1
    arr[i] = arr[min]; — 1
    arr[min] = temp; — 1
    y
}
T(n) = n(n-1)
      = n2 - 2n + 1
      ~ O(n2)
```

average case :

```
for (i=0; i<n-1; i++) — n-1
{
    min=i; — 1
    for (j=i+1; j<n; j++) — n
    {
        if (arr[j]<arr[min]) — 1
        {
            min=j; — 1
        }
        y
    }
    temp = arr[i]; — 1
    arr[i] = arr[min]; — 1
    arr[min] = temp; — 1
    y
}
T(n) = n(n-1)
      = n2 - 2n + 1
      ~ O(n2)
```



worst case :

```
for (i=0; i<n-1; i++) — n-1
{
    min=i; — 1
    for (j=i+1; j<n; j++) — n
    {
        if (arr[j]<arr[min]) — 1
        {
            min=j; — 1
        }
        temp = arr[i]; — 1
        arr[i] = arr[min]; — 1
        arr[min] = temp; — 1
    }
    T(n) = n(n-1)
        = n2 - 2n + 1
        = O(n2)
}
```

**MERGE SORT ALGORITHM :**

4) Merge sort :-  
Input : Array A with n elements  
Output : Sorted array A in ascending order

- 1) start
- 2) Read n and array A
- 3) Divide the array into two halves
- 4) Recursively divide each half until single elements remain
- 5) compare elements of the two halves
- 6) merge them in sorted order
- 7) Repeat the merging until the complete array is sorted
- 8) Display sorted array A
- 9) stop.

PROGRAM:

```
#include <iostream> using
namespace std;

class mergesort
{ public:
    int size, arr[100], i;

    void arrayinput()
    {      cout << "Enter array
size: ";      cin >> size;
        cout << "Enter array elements:\n";
        for (i = 0; i < size; i++)      {      cin >> arr[i];
        }
    }

    void printarray()
    {      for (i = 0; i < size;
i++)
        {      cout <<
arr[i] << " ";
        }      cout
<< endl;
    }

    void merge(int low, int mid, int high)
    {      int temp[100];      int i =
low, j = mid + 1, k = low;

        while (i <= mid && j <= high)
        {      if (arr[i]
<= arr[j])
        {
temp[k] = arr[i];
i++;
        }      else
        {      temp[k] =
arr[j];      j++;
        }      k++;
    }
```



```
    }

    while (i <= mid)
    {
        temp[k]
= arr[i];
        i++;
        k++;
    }

    while (j <= high)
    {
        temp[k] = arr[j];    j++;    k++;
    }

    for (i = low; i <= high; i++)
    {
        arr[i] =
temp[i];
    }
}

void mergellogic(int low, int high)
{
    if (low <
high)
    {
        int mid = (low +
high) / 2;

        mergellogic(low, mid);
        mergellogic(mid + 1, high);

        merge(low, mid, high);
    }
}

};

int main() {
mergesort ms;

    ms.arrayinput();

    cout << "Before sorting: ";    ms.printarray();
}
```

---

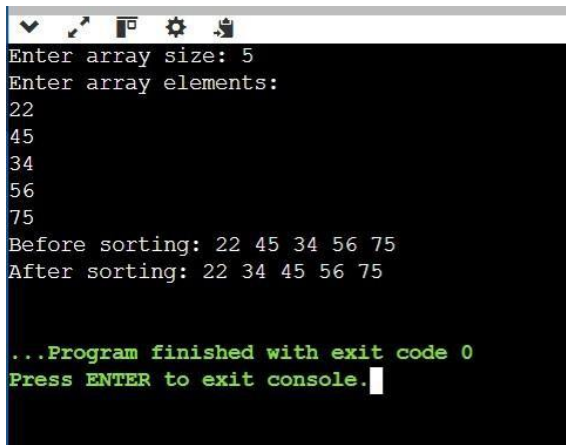
```

ms.mergelogsic(0, ms.size - 1);

cout << "After sorting: ";
ms.printarray();

return 0; }
OUTPUT :

```



```

Enter array size: 5
Enter array elements:
22
45
34
56
75
Before sorting: 22 45 34 56 75
After sorting: 22 34 45 56 75

...Program finished with exit code 0
Press ENTER to exit console.

```

time complexity : Step Count / Frequency Count Method )

best case & average case & worst case :

Merge Sort :-

```

void merge (int low, int mid, int high)
{
    int temp[100];
    int i = low, j = mid + 1, k = low;
    while (i <= mid && j <= high) {
        if (arr[i] <= arr[j]) {
            temp[k] = arr[i];
            i++;
        }
        else {
            temp[k] = arr[j];
            j++;
        }
        k++;
    }
}

```

```

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  Sub (a) in (b)
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  = 16T(n/16)+2n+2n
  = 16T(n/16)+4n
  T(n) = 16T(n/16)+4n

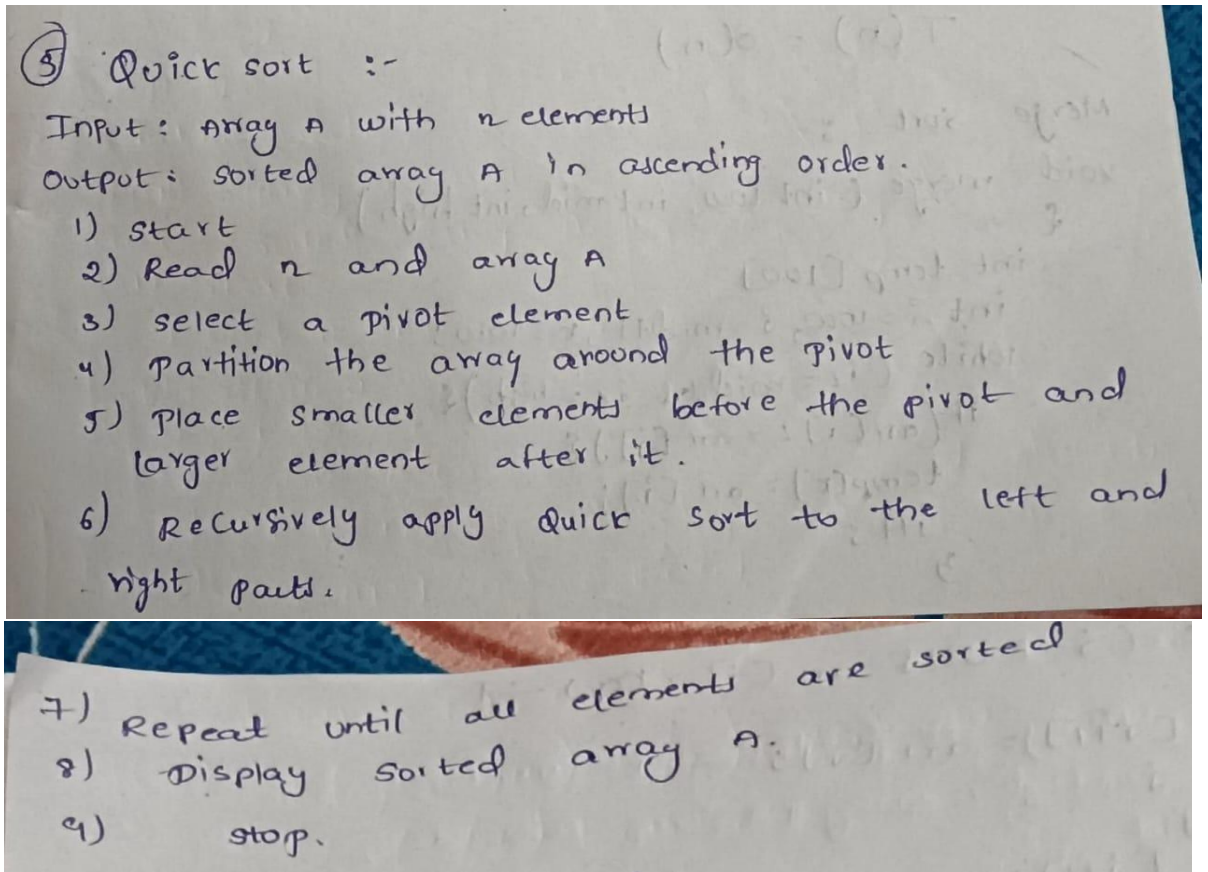
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  'IC,-..)» (1)+ nl°C),...,
  ,C, ....:) =- r) 100 ,.

```





### QUICK SORT ALGORITHM:



### PROGRAM :

```
#include <iostream> #include  
<utility> using  
namespace std;
```

```
class QuickSort {  
public: int size,  
arr[100];
```

```
    void arrayinput()  
    {        cout << "Enter array  
size: ";    cin >> size;  
  
        cout << "Enter array elements: ";  
        .....for (int i = 0; i < size; i++)
```



```
        {            cin
>> arr[i];
        }
    }

    void printarray()
    {            for (int i = 0; i < size;
i++)
        {            cout <<
arr[i] << " ";
        }            cout
<< endl;
    }

    int partition(int lb, int ub)
    {            int pivot =
arr[lb];
    int start = lb;
        int end = ub;

        while (start < end)
        {            while (start <= ub && arr[start]
<= pivot)                start++;

            while (arr[end] > pivot)
end--;
            if (start <
end)
                swap(arr[start], arr[end]);
        }

        swap(arr[lb], arr[end]);
    return end;
    }

    void quicksortlogic(int lb, int ub)
    {            if (lb < ub)            {
int loc = partition(lb, ub);
```

---

```
quicksortlogic(lb, loc - 1);  
quicksortlogic(loc + 1, ub);  
    }  
}
```

```
void sort() {  
quicksortlogic(0, size - 1);  
}  
};
```

```
int main()  
{
```

```
    QuickSort qs;
```

```
    qs.arrayinput();
```

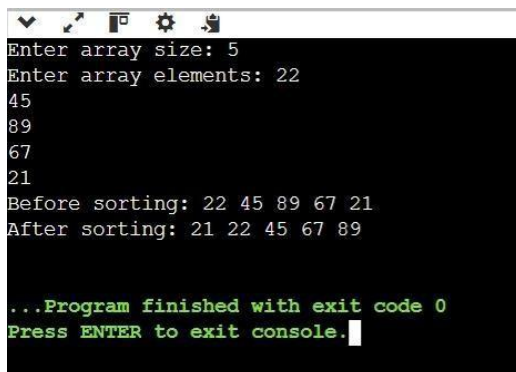
```
    cout << "Before sorting: ";  
    qs.printarray();
```

```
    qs.sort();
```

```
    cout << "After sorting: ";  
    qs.printarray();
```

```
    return 0;  
}
```

OUTPUT :



```
Enter array size: 5  
Enter array elements: 22  
45  
89  
67  
21  
Before sorting: 22 45 89 67 21  
After sorting: 21 22 45 67 89  
  
...Program finished with exit code 0  
Press ENTER to exit console.
```

TIME COMPLEXITY : Step Count / Frequency Count Method ) best case :

Recurrence Relation

$$T(n) = 2T(n/2) + O(n)$$

expand

$$T(n) = 2(2T(n/4) + n/2) + n$$

$$T(n) = 4T(n/4) + 2n$$

again

$$T(n) = 8T(n/8) + 3n$$

$$= 16T(n/16) + 4n$$

$$T(n) = 2^k T(n/2^k) + kn$$

$n = 2^k$

add log on both sides

$$T(n) = 2 \log n (1) + \log n (n)$$

$$T(n) = n \log^2 + n \log n$$

$$T(n) = n(1) + n \log n$$

$$T(n) = n \log n$$

average case :

Recurrence Relation

$$T(n) = 2T(n/2) + O(n)$$

expand

$$T(n) = 2(2T(n/4) + n/2) + n$$

$$T(n) = 4T(n/4) + 2n$$

again

$$T(n) = 8T(n/8) + 3n$$

$$= 16T(n/16) + 4n$$

$$T(n) = 2^k T(n/2^k) + kn$$

$n = 2^k$

add log on both sides

$$T(n) = 2 \log n (1) + \log n (n)$$

$$T(n) = n \log^2 + n \log n$$

$$T(n) = n(1) + n \log n$$

$$T(n) = n \log n$$

worst case :

